FCC-ue & Muon-Collider $4b$ Production

28 October 2026

Process inventory | completed channels and two polarized extensions

LO / parton level

Scope. Process definitions only; no cross sections. **CC** denotes neutrino final states; **NC** denotes outgoing charged leptons.

01 Original charged-current study — COMPLETED

	FCC-ue: μ^+e^-	Muon collider: $\mu^+\mu^-$
A. Double-Higgs	$\mu^+e^- \to \nu_e\bar{\nu}_\mu HH$ $\to \nu_e\bar{\nu}_\mu\, b\bar{b}\, b\bar{b}$	$\mu^+\mu^- \to \nu_\ell\bar{\nu}_\ell HH$ $\to \nu_\ell\bar{\nu}_\ell\, b\bar{b}\, b\bar{b}$
B. Complete CC $4b$	$\mu^+e^- \to \nu_e\bar{\nu}_\mu\, b\bar{b}\, b\bar{b}$	$\mu^+\mu^- \to \nu_\ell\bar{\nu}_\ell\, b\bar{b}\, b\bar{b}$

For the muon collider, $\ell = e, \mu, \tau$ (flavor channels summed). HH samples use on-shell Higgs production followed by two $H \to b\bar{b}$ decays. The complete CC-$4b$ amplitudes include all allowed Higgs-mediated and nonresonant diagrams, with their interference. **Benchmarks evaluated:** FCC-ue $E_{\mu^+}/E_{e^-} = 2000/120$ and $5000/183$ GeV; MuC $E_{\mu^+}/E_{\mu^-} = 500/500$ and $1000/1000$ GeV; both unpolarized and $P_{\mu^+} = +80\%$.

02 Additional neutral-current study: FCC-ue, unpolarized 1 TeV — COMPLETED

NC double-Higgs signal	$\mu^+e^- \to \mu^+e^- HH \to \mu^+e^-\, b\bar{b}\, b\bar{b}$
NC associated Higgs	$\mu^+e^- \to \mu^+e^-\, b\bar{b}H \to \mu^+e^-\, b\bar{b}\, b\bar{b}$
NC $4b$ continuum	$\mu^+e^- \to \mu^+e^-\, b\bar{b}\, b\bar{b}$ (no internal H)

FCC-ue beams: $E_{\mu^+} = 2000$ GeV, $E_{e^-} = 120$ GeV; both unpolarized. The continuum is defined by omitting internal Higgs diagrams (`/ h` in MadGraph), so this is a diagram-defined estimate, not automatically a gauge-invariant separation.

03 Two requested polarized NC associated-Higgs extensions — NEXT / PENDING

FCC-ue	$\mu^+e^- \to \mu^+e^-\, b\bar{b}H \to \mu^+e^-\, b\bar{b}\, b\bar{b}$ $E_{\mu^+} = 5000$ GeV; $E_{e^-} = 183$ GeV; $P_{\mu^+} = +80\%$, $P_{e^-} = 0$.
Muon collider	$\mu^+\mu^- \to \mu^+\mu^-\, b\bar{b}H \to \mu^+\mu^-\, b\bar{b}\, b\bar{b}$ $E_{\mu^+} = E_{\mu^-} = 1000$ GeV ($\sqrt{s} = 2$ TeV); $P_{\mu^+} = +80\%$, $P_{\mu^-} = 0$.

04 What was missing from the original study?

Original CC final state: $\mu^+e^- \to \nu_e\bar{\nu}_\mu\, b\bar{b}\, b\bar{b}$. Its complete amplitude already includes allowed Higgs-mediated diagrams for *this neutrino final state*.

Additional NC channel: $\mu^+e^- \to \mu^+e^- b\bar{b}H \to \mu^+e^-\, b\bar{b}\, b\bar{b}$. This has *outgoing charged leptons* and was not included in the original CC study. The CC and NC external final states do not interfere.